CUSTOMER EXPERIMENT / PLANNING

Experiment Design Package

Credit line test - illustrative

SYNTHETIC DEMO | The data and results in this package are illustrative.

01 / Sample size recommendation

Use these targets to plan random assignment. The total includes the Control and every Test option.

Role	Option	Analyzable N	Required N	Allocation
Control	Credit line 1000	1,080	1,080	32.3%
Test	Credit line 2000	1,060	1,060	31.7%
Test	Credit line 3000	1,206	1,206	36.0%
Total	All options	3,346	3,346	100.0%

The shared Control (Credit line 1000) needs 1,080 customers. Required N includes any configured reserve for missing outcomes.

This is a prospective sample-size estimate, not a result or an approval to launch.

02 / Design at a glance

Design item	Recorded setting
Primary metric	Revolving Balance
Minimum detectable effect	5% of the Control baseline
Group by	FICO group
Partners	Holland, PriceLine
Selected groups	FICO 650–699, FICO 700–749, FICO 750+
Control	Credit line 1000
Target power / significance	80% per comparison / 5.0% nominal alpha

The observation window and metric definitions must be confirmed with the data owner if not supplied. No eligibility or recruitment timeline is inferred.

03 / Statistical settings and assumptions

Setting	Value / interpretation
Experiment owner	Not provided
Data source	Synthetic demo
Calculation time (UTC)	2026-10-27T19:44:54+00:00
Randomization unit	Customer
Decision scope	Overall
Allocation	Pairwise Neyman with a shared Control
Missing-outcome reserve	0.0%
Multiple comparison protection	On; family size 4; alpha per comparison 0.0125
Target power	80.0% for each planned comparison
Guardrail	NCL (included in plan; margin 4)

What the plan assumes

The historical Control counts define the Partner and group weights. The experiment must maintain that same mix in Control and Test options.

The variance inputs are historical estimates. Historical differences between strategies are not interpreted as causal effects; the MDE is the business-specified target.

Power is a per-comparison target, not the probability that all Tests or metrics succeed together. The calculation does not fully incorporate uncertainty in historical moments.

The design uses fixed-horizon inference. Repeated early significance checks require a separately specified method. Confirm randomization, assignment integrity and observation-window maturity before analysis.

04 / Calculation verification

Each Test is compared with the same Control. The final integer allocation has been checked against the selected target power.

Test option	Initial control N	Initial test N	Starting ratio
Credit line 2000	1,038	1,099	1.058
Credit line 3000	1,080	1,206	1.116

Metric / test	Control N	Test N	Difference*	Power
Revolving Balance / Credit line 2000	1,080	1,060	59.1224	92.3%
Revolving Balance / Credit line 3000	1,080	1,206	59.1224	92.6%
NCL / Credit line 2000	1,080	1,060	4	80.0%
NCL / Credit line 3000	1,080	1,206	4	80.0%

* Difference is the planned detectable effect or guardrail tolerance, expressed in metric units. These are model assumptions, not observed effects.

05 / Calculation formulas

Notation is shown for the continuous-metric planning model used in this package.

A. Combine non-overlapping historical groups

$$N = \Sigma\, n_i \quad \text{Mean} = \Sigma\,(n_i \times \text{mean}_i) / N$$
$$\text{Variance} = \Sigma\,[(n_i - 1) \times \text{variance}_i + n_i \times (\text{mean}_i - \text{Mean})^2] / (N - 1)$$

n is historical customer count; the second variance term captures differences between historical group means.

B. Define target population and effect

$$w_h = \text{Control historical } N_h \,/\, \text{total Control historical } N$$
$$\text{Baseline} = \Sigma\,(w_h \times \text{Control mean}_h)$$
$$\text{Relative effect in metric units} = |\text{Baseline}| \times \text{MDE (\%)} / 100$$

h denotes a Partner and group. The user-specified absolute MDE is used directly.

C. Variance and Welch-Satterthwaite degrees of freedom

$$V = \Sigma\,[w_h^2 \times (s_{\text{Control},h}^2 / n_{\text{Control},h} + s_{\text{Test},h}^2 / n_{\text{Test},h})]$$
$$df = V^2 / \Sigma\,[v_j^2 / (n_j - 1)]$$

s^2 is historical variance, n is the planned analyzable sample and each v is one variance component in V.

D. Neyman ratio and shared Control

$$\text{Test / Control starting ratio} = \sqrt{(\Sigma\, w_h \times s_{\text{Test},h}^2)} \,/\, \sqrt{(\Sigma\, w_h \times s_{\text{Control},h}^2)}$$

Size each Test against Control; keep the largest Control requirement, then recalculate each Test and verify the final integer sample sizes.

E. Power and comparison protection

$$\text{Noncentrality} = \text{effect} / \sqrt{V}$$
$$\text{Adjusted alpha} = \text{alpha} / \text{number of comparisons in the family}$$

A noncentral t distribution is used for primary-metric power. The alpha adjustment applies only when protection is enabled. Guardrail planning uses its margin and one-sided non-inferiority power with a conservative critical value.

F. Missing-outcome reserve

$$\text{Recruitment target} = \text{round up}\,[\text{analyzable sample} / (1 - \text{missing-outcome fraction})]$$

The reserve is applied separately to each Partner, group and option.

06 / Population mix

Weights are based on historical Control sample counts and are shared across experiment options.

Partner	FICO group	Weight
Holland	FICO 650–699	14.97%
Holland	FICO 700–749	16.67%
Holland	FICO 750+	18.37%
PriceLine	FICO 650–699	14.97%
PriceLine	FICO 700–749	16.67%
PriceLine	FICO 750+	18.37%

07 / Historical inputs

These are the pooled statistics used by the calculation, not a copy of individual customer records.

Partner / group	Option	Metric	Mean	Variance	N
Holland / FICO 650–699	Credit line 1000	Revolving Balance	1,015.15152	100,500.736	660
Holland / FICO 650–699	Credit line 1000	NCL	45.6060606	608.152619	660
Holland / FICO 650–699	Credit line 2000	Revolving Balance	1,115.46099	113,284.198	705
Holland / FICO 650–699	Credit line 2000	NCL	50.5673759	704.169105	705
Holland / FICO 650–699	Credit line 3000	Revolving Balance	1,215.73333	126,867.352	750
Holland / FICO 650–699	Credit line 3000	NCL	55.5333333	808.159769	750
Holland / FICO 700–749	Credit line 1000	Revolving Balance	1,115.64626	113,282.926	735
Holland / FICO 700–749	Credit line 1000	NCL	41.5442177	703.915958	735
Holland / FICO 700–749	Credit line 2000	Revolving Balance	1,215.89744	126,866.716	780
Holland / FICO 700–749	Credit line 2000	NCL	46.5128205	807.920246	780
Holland / FICO 700–749	Credit line 3000	Revolving Balance	1,316.12121	141,250.228	825
Holland / FICO 700–749	Credit line 3000	NCL	51.4848485	919.902986	825
Holland / FICO 750+	Credit line 1000	Revolving Balance	1,216.04938	126,866.079	810
Holland / FICO 750+	Credit line 1000	NCL	37.4938272	807.697737	810
Holland / FICO 750+	Credit line 2000	Revolving Balance	1,316.25731	141,250.144	855
Holland / FICO 750+	Credit line 2000	NCL	42.4678363	919.69188	855
Holland / FICO 750+	Credit line 3000	Revolving Balance	1,416.44444	156,433.951	900
Holland / FICO 750+	Credit line 3000	NCL	47.4444444	1,039.66766	900
PriceLine / FICO 650–699	Credit line 1000	Revolving Balance	1,135.15152	100,500.736	660

Partner / group	Option	Metric	Mean	Variance	N
PriceLine / FICO 650–699	Credit line 1000	NCL	48.0060606	608.152619	660
PriceLine / FICO 650–699	Credit line 2000	Revolving Balance	1,235.46099	113,284.198	705
PriceLine / FICO 650–699	Credit line 2000	NCL	52.9673759	704.169105	705
PriceLine / FICO 650–699	Credit line 3000	Revolving Balance	1,335.73333	126,867.352	750
PriceLine / FICO 650–699	Credit line 3000	NCL	57.9333333	808.159769	750
PriceLine / FICO 700–749	Credit line 1000	Revolving Balance	1,235.64626	113,282.926	735
PriceLine / FICO 700–749	Credit line 1000	NCL	43.9442177	703.915958	735
PriceLine / FICO 700–749	Credit line 2000	Revolving Balance	1,335.89744	126,866.716	780
PriceLine / FICO 700–749	Credit line 2000	NCL	48.9128205	807.920246	780
PriceLine / FICO 700–749	Credit line 3000	Revolving Balance	1,436.12121	141,250.228	825
PriceLine / FICO 700–749	Credit line 3000	NCL	53.8848485	919.902986	825
PriceLine / FICO 750+	Credit line 1000	Revolving Balance	1,336.04938	126,866.079	810
PriceLine / FICO 750+	Credit line 1000	NCL	39.8938272	807.697737	810
PriceLine / FICO 750+	Credit line 2000	Revolving Balance	1,436.25731	141,250.144	855
PriceLine / FICO 750+	Credit line 2000	NCL	44.8678363	919.69188	855
PriceLine / FICO 750+	Credit line 3000	Revolving Balance	1,536.44444	156,433.951	900
PriceLine / FICO 750+	Credit line 3000	NCL	49.8444444	1,039.66766	900

08 / Reproducibility and review notes

Experiment ID: 32062dc6-bd29-44c0-bcf2-7fc0dd4fd4c1

Historical input fingerprint: 604240cec6c2267944b1a39e132ec857f00a98a9036ce4898d9e692246d9a507

Statistical engine code identifier: customer-v2-sha256:22a48312d3977f1fa7ce93127e4d1d4909fe104bb255b0eb94104a9d03f2cf0a

Retain the original source file, data definitions and application version in an approved location. The engine identifier does not cover the interface or input adapters. This package records calculation inputs and assumptions; it does not verify the source, randomization, review or approval.